

WallBo: A Social Robot for Fostering Hand Hygiene For Children

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Abstract

Hand hygiene is a critical public health intervention, yet habit formation remains elusive among children in resource-constrained environments. We present WallBo, a low-cost social robot designed to teach WHO handwashing techniques. Building upon our previous cross-cultural study which utilised teleoperation and identified a sharp “novelty drop-off” in engagement, this iteration features upgraded interaction and hardware, a refined physical design, and fully autonomous operation. To sustain long-term handwashing compliance, WallBo integrates three novel mechanisms: (1) Augmented Reality Gamification with “Virtual Germs” to bridge the gap between abstract infection risk and concrete preventive action; (2) Velocity-Adaptive Verbal Scaffolding for real-time sensorimotor feedback; and (3) Optical Flow fallback for occlusion handling. Unlike cloud-dependent systems, WallBo operates entirely offline on edge hardware, ensuring privacy by design. This demonstration allows attendees to interact with WallBo’s adaptive logic, experiencing how embodied, autonomous AI can bridge gaps in motor skills and learning development in the handwashing context.

CCS Concepts

• Computer systems organization → Robotics; • Human-centered computing → Interaction design; User studies; Mixed / augmented reality; • Computing methodologies → Computer vision.

Keywords

Social Robotics; Hand Hygiene; Edge Computing; Child-Robot Interaction; Public Health; Privacy by Design; Augmented Reality

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1 Introduction

Infectious diseases pose significant public health challenges globally, particularly in low-resource settings where limited sanitation intensifies the risk. In this context, effective hand hygiene is recognised by the World Health Organization (WHO) as the single most effective measure to prevent disease transmission [21]. Despite its importance, adherence to proper handwashing remains critically low, particularly among young children who often wet their hands without using soap or performing the necessary scrubbing steps [11]. This non-compliance contributes to estimated millions of missed school days and preventable deaths annually [17, 20].

As noted by Hussam et al. [12], the challenge is not merely providing information, but forming lasting habits through engaging environments [5]. Socially Assistive Robots (SARs) have shown promise in addressing this by leveraging physical presence and interactive capabilities to make health education memorable [1]. However, the widespread deployment of such systems is often hindered by critical barriers: (1) high hardware costs, (2) reliance on high-bandwidth internet for cloud processing, (3) significant privacy concerns regarding video capture in sensitive spaces like washrooms, and (4) a lack of sustained long-term engagement.

In this demonstration, we introduce an iteration of “WallBo”, a robust, privacy-preserving robotic system designed for autonomous operation in school washrooms. This system represents a significant technical evolution from previous prototypes, incorporating advanced Augmented Reality (AR) and adaptive audio feedback to bridge the gap between education and sustained habit formation.

2 Robot Design

We are developing the WallBo robot as a low-cost (USD ≈500) portable robotic platform with minimal expressive capabilities [8]. According to Bartneck and Forlizzi [2], the physical characteristics of a social robot, such as shape, size, and material, should align with the task to prevent creating false expectations. Therefore, WallBo was designed in the shape of a hand to symbolically reinforce the theme of handwashing. The bright green colour of the robot was chosen to represent good health and wellbeing.

2.1 Physical and Social Cues

WallBo measures 35 cm high and 30 cm wide (See fig. 1d). It is equipped with animated eyes designed to enhance the “being-watched” or “Hawthorne effect” [13]; previous research by Pfattheicher et al. [15] showed that even subtle cues, like images of eyes,

can significantly improve hand hygiene compliance in public settings. The robot features animated eyes and mouth displayed on a small screen that renders expressive behaviours while providing hand washing instructions to the user. WallBo is constructed using off-the-shelf electronics housed within custom-designed 3D-printed components, ensuring reproducibility and durability in school environments.

2.2 Voice and Educational Content

The interaction using WallBo is designed to be educational and engaging. The robot’s speech is tailored to the specific cultural context, using a child-friendly voice with a localised language and accent. The pre-generated utterances are processed to lower the pitch with the goal of making the voice gender-neutral and avoiding potential gender biases. Previous research indicates that such voices effectively capture children’s attention in child-robot interaction studies [18]. To facilitate learning, WallBo displays animations of the key steps of proper handwashing on its screen while providing verbal step-by-step instructions to one user at a time. The animation depicts the proper handwashing technique synchronised with the verbal instructions, ensuring that the educational content remains accessible to children with varying literacy levels. Notably, the robot does not utilise speech recognition; it responds to the user solely based on detected hand movements.



Figure 1: Field deployment in Indian and UK schools showing WallBo v1.0 providing instructions (a, b); WallBo v1.0 used in WoZ study (c); WallBo v2.0 autonomous system (d).

3 Background and Motivation

3.1 Findings from Previous Deployment

Our prior work evaluated robot-assisted hand hygiene interventions (one-week deployment) across diverse socio-economic contexts [6], specifically in schools in India [7] and the UK [8] ($N = 77$, see Fig. 1a, 1b). The results highlighted critical disparities driven by socio-economic status and cultural context:

- **The Novelty Effect and Drop-off:** In lower socio-economic settings (India), children initially demonstrated high engagement and learning gains. However, longitudinal data revealed a sharp “novelty drop-off” once the initial excitement of the robot faded, leading to reduction in handwashing compliance.
- **Baseline Knowledge vs. Engagement:** In contrast, children in higher-income settings (UK) possessed higher baseline hygiene knowledge and technology familiarity. For these cohorts, static instructions were perceived as repetitive, leading to quicker disengagement.
- **Infrastructure Constraints:** The study also exposed the fragility of cloud-dependent systems in under-resourced schools. Intermittent internet connectivity often rendered cloud processing unreliable, necessitating a move toward fully offline, edge-based architectures.

3.2 Motivation for WallBo 2.0

The findings from our previous study (section 3.1) suggest that a static, “one-size-fits-all” intervention is insufficient for sustaining long-term behaviour change. To bridge the gap between initial novelty and permanent habit formation, a system must provide **continuous, adaptive reinforcement**. While our previous field study [6] provided critical insights into cross-cultural engagement, it relied on a Wizard of Oz (WoZ) protocol where a human teleoperator controlled the robot’s responses. Although effective for initial validation and testing, this approach is not scalable or sustainable for long-term deployment in schools.

The current system represents a fully autonomous upgrade. By migrating the logic to an onboard Edge computer (NVIDIA Jetson Nano Orin), WallBo now operates independently of human oversight or external internet connectivity. Furthermore, the physical design has been iteratively refined based on the feedback from children and school authorities collected during the initial trials. The new design (see Fig. 1d WallBo 2.0) is constructed from robust, modular 3D-printed components, addressing the school authorities’ request for durability and ease of maintenance, while the refined aesthetic incorporates specific design cues preferred by the children to enhance the robot’s appearance. Also, the current system now moves beyond passive instruction to active, gamified engagement (via AR and Adaptive Feedback) to help maintain user interest long after the novelty has worn off.

4 System Design and Architecture

Based on the constraints identified above, WallBo acts as a “situated intervention” [16], a portable unit near the sink/handwashing station to provide “Just-in-Time” prompts. This design philosophy addresses a fundamental challenge in health behaviour change: the gap between knowing what to do and actually doing it in the

appropriate context. By embedding the intervention directly where handwashing occurs, WallBo eliminates the cognitive burden of remembering and transferring classroom-taught protocols to real-world application. **The system's portability** allows it to be deployed flexibly across multiple handwashing stations within a school, while its autonomous operation ensures consistent reinforcement without requiring continuous teacher supervision or infrastructure modifications.

4.1 Hardware and Edge Architecture

WallBo is a stand-alone unit comprising a screen, an RGB camera, and speaker, powered by an Edge AI Computer. The system architecture follows a strict *Edge AI* paradigm: all processing occurs locally, ensuring no interaction data leaves the device (See Fig. 2).

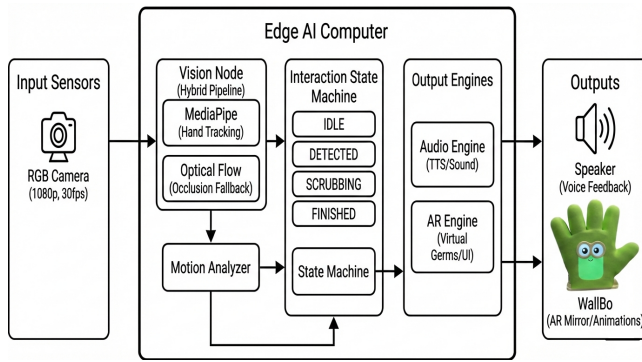


Figure 2: System Architecture: The Vision Node utilizing a hybrid pipeline of Hand Tracking (MediaPipe [14]) and Optical Flow. The central State Machine orchestrates the Audio and AR Engines to deliver velocity-adaptive feedback and virtual germ overlays.

4.2 Hybrid Logic Algorithm

The core innovation of WallBo is its ability to handle sensor uncertainty (occlusion) while maintaining user engagement. Algorithm 1 details the execution loop that arbitrates between hand tracking and optical flow, ensuring the robot remains responsive even when hands are covered in soap foam or occluded.

4.3 Interaction Design

This iteration of WallBo integrates three novel features, each grounded in behavioural science literature to address the “novelty drop-off”. Rather than relying solely on the robot’s physical presence, these features create dynamic, personalised interactions that provide continuous value beyond initial curiosity.

4.3.1 AR “Virtual Germs” (Cognitive Gap Reduction). : The *Health Belief Model* [3] posits that preventative action is driven by perceived susceptibility and severity. However, germs are invisible, making the risk abstract for children. We employ *Augmented Reality* to visualize this invisible threat. Using *landmark interpolation*, the system renders procedurally animated “viral nodes” anchored to the metacarpal bones of the user’s hands. These nodes scale and

Algorithm 1 WallBo Hybrid Interaction Loop- Pseudo Code

```

while System Active do
  Input  $\leftarrow$  GetFrame(Camera)
  if Hands Detected then                                 $\triangleright$  Primary Mode
    RenderAR(VirtualGerms)
    if Germs Touched then Score  $\leftarrow$  Score + 1
    end if
     $v \leftarrow$  CalcWristVelocity()
    AudioFeedback  $\leftarrow$  GETVELOCITYPROMPT( $v$ )  $\triangleright$  "Faster" /
    "Slow Down"
  else                                                     $\triangleright$  Occlusion Fallback (Soap/Foam)
    Motion  $\leftarrow$  OpticalFlow(Input)
    if Motion > Threshold then
      Score  $\leftarrow$  Score + CompensationRate  $\triangleright$  Prevent user
      frustration
    end if
  end if
  if Score  $\geq$  Target then ADVANCELEVEL
  end if
end while

```

rotate with the hand, creating a convincing illusion of contamination. Previous research suggests that visualising contamination (e.g., using UV gels) significantly improves hygiene compliance by bridging the gap between abstract health risks and concrete sensory experience [11]. WallBo digitises this proven method, using AR to provide immediate operant conditioning: the user scrubs, and the “germs” visually pop (positive reinforcement).

4.3.2 Velocity-Adaptive Verbal Scaffolding (Motor Learning). : While visual feedback is effective, auditory feedback is superior for regulating temporal aspects of motor tasks. This is known as *Concurrent Augmented Feedback* [19]. WallBo calculates the velocity of the user’s wrist movements ($v = \Delta d / \Delta t$) and maps this data to the robot’s vocal feedback in real-time. If v drops below a threshold, the robot urges “Faster!”; if v exceeds control limits, it cautions “Slow down.” Effenberg et al. [9] demonstrated that the sonification of movement parameters enhances sensorimotor learning more effectively than visual feedback alone, as it reduces visual cognitive load while providing high-resolution temporal cues. WallBo applies this principle through adaptive verbal scaffolding to guide proper scrubbing rhythm.

4.3.3 Optical Flow Fallback (System Trust). : Inconsistent system performance erodes user trust and engagement. A key challenge in hand movement monitoring is *texture occlusion*: soap foam/hands can obliterate the edge features required by hand trackers such as MediaPipe [14]. We implement a hybrid vision pipeline (see Algorithm 1). When hand tracking confidence drops, the system seamlessly switches to a dense *Optical Flow* algorithm [10]. It measures pixel motion intensity in the region of interest; if high-velocity motion is detected, the system infers that scrubbing is continuing. This approach aligns with robust action recognition frameworks for occluded environments [4], ensuring the user is never “punished” (by a stalled progress bar) for using too much soap.

5 The Demonstration Experience

Attendees will engage with WallBo in a simulated washroom environment. **1. Approach & Greeting:** As the user approaches, WallBo's vision system wakes from IDLE. The robot greets the user: "Hi there! I'm your washing buddy! Show me your hands!" **2. The Washing Game (AR):** The screen displays a real-time mirror feed. Virtual "germs" will appear on the attendee's hands. The robot challenges them to remove the germs. **3. Adaptive Feedback:** Attendees are encouraged to vary their scrubbing speed. They will hear the robot's voice react instantly to their tempo. **4. Victory:** Upon completion, the system triggers a "High Five" audio cue and visual celebration. Interaction time varies between 1–2 minutes, based on the attendee's pace through the handwashing steps.

6 Relevance to CHI Community

This demonstration contributes to the CHI community by exemplifying how Human–Robot Interaction (HRI) and Computer Vision can be scalably deployed in resource-constrained, real-world settings. By presenting this "privacy-by-design" architecture, we offer a transferable framework for ubiquitous sensing systems that prioritises the protection of vulnerable populations (children) while delivering scientifically grounded health interventions.

7 Discussion and Future Work

7.1 Limitations

While the transition to Edge AI mitigates latency and privacy concerns, several limitations remain. First, **environmental robustness** is challenged by extreme lighting conditions common in school washrooms, which can degrade the 30 FPS throughput. Second, the current computer vision pipeline assumes a **single-user interaction** model; crowded washroom scenarios may introduce tracking ambiguity. Finally, a significant pedagogical limitation is that the system currently does not validate adherence to the specific World Health Organization (WHO) handwashing steps (e.g., interlacing fingers, rotational rubbing). The intervention currently functions as a motivational tool for general scrubbing intensity and duration, rather than a technical tutor for precise handwashing mechanics.

7.2 Future Work

Our immediate focus is a **longitudinal deployment** (12 weeks) to evaluate habit retention post-intervention. Future technical iterations will incorporate **multi-modal analysis** to distinguish active users in noisy environments. Crucially, we aim to advance the computer vision pipeline to **classify distinct handwashing gestures**. This would enable the system to guide children through the complete WHO protocol, verifying each specific gesture (e.g., palm-to-palm vs. back-of-hand) in real-time rather than relying solely on generic motion analysis.

7.3 Ethical Implications

We have adopted a rigorous **privacy-by-design** framework. All visual data is processed on the Edge AI unit; frames are analysed for interaction states and immediately discarded. Furthermore, to address potential **algorithmic bias** in hand tracking across diverse skin tones, we are committed to continuous auditing of our hand

tracking implementation to ensure equitable feedback for children from all demographic backgrounds.

8 Sustainability Statement

In alignment with CHI 2026 sustainability goals, this demonstration utilises a low-power, edge-based hardware architecture (NVIDIA Jetson Orin Nano, < 15 Watt), rather than requiring energy-intensive remote GPU servers or cloud-based solutions. By democratising hand hygiene education in resource-constrained settings, WallBo addresses UN Sustainable Development Goals 3 (Good Health), 4 (Quality Education), and 6 (Sanitation), offering a potential scalable and accessible solution that can reduce disease transmission and educational inequality without relying on expensive or energy-intensive infrastructure. This approach contributes to the creation of more sustainable health systems and a future where HCI technology can actively help to bridge gaps in healthcare education.

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